QuantEdge
Let's Make Quant Accessible To All

QUANT FINANCE

MARKET RESEARCH REPORT

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Introduction

Quantitative Finance, often referred to as “Quant Finance,” is a discipline that applies mathematical models, statistical techniques, and computational algorithms to analyze financial markets and securities. It encompasses various areas, including risk management, derivative pricing, portfolio optimisation, and algorithmic trading.

The roots of Quantitative Finance trace back to the 1950s with Harry Markowitz’s Modern Portfolio Theory, which introduced the concept of portfolio diversification to optimize returns relative to risk. Subsequent developments, such as the Black-Scholes model for option pricing and advancements in computing power, have significantly expanded the field’s scope.

Key Components

- **Mathematical Modelling:** Utilising stochastic calculus, differential equations, and linear algebra to model financial instruments and market behaviours. E.g., derivatives pricing, risk metrics, portfolio optimisation
- **Statistical Analysis:** Employing statistical methods to identify patterns, correlations, and trends within financial data.
- **Computational Algorithms:** Developing algorithms for tasks such as high-frequency trading, risk assessment, and automated portfolio management.

Applications

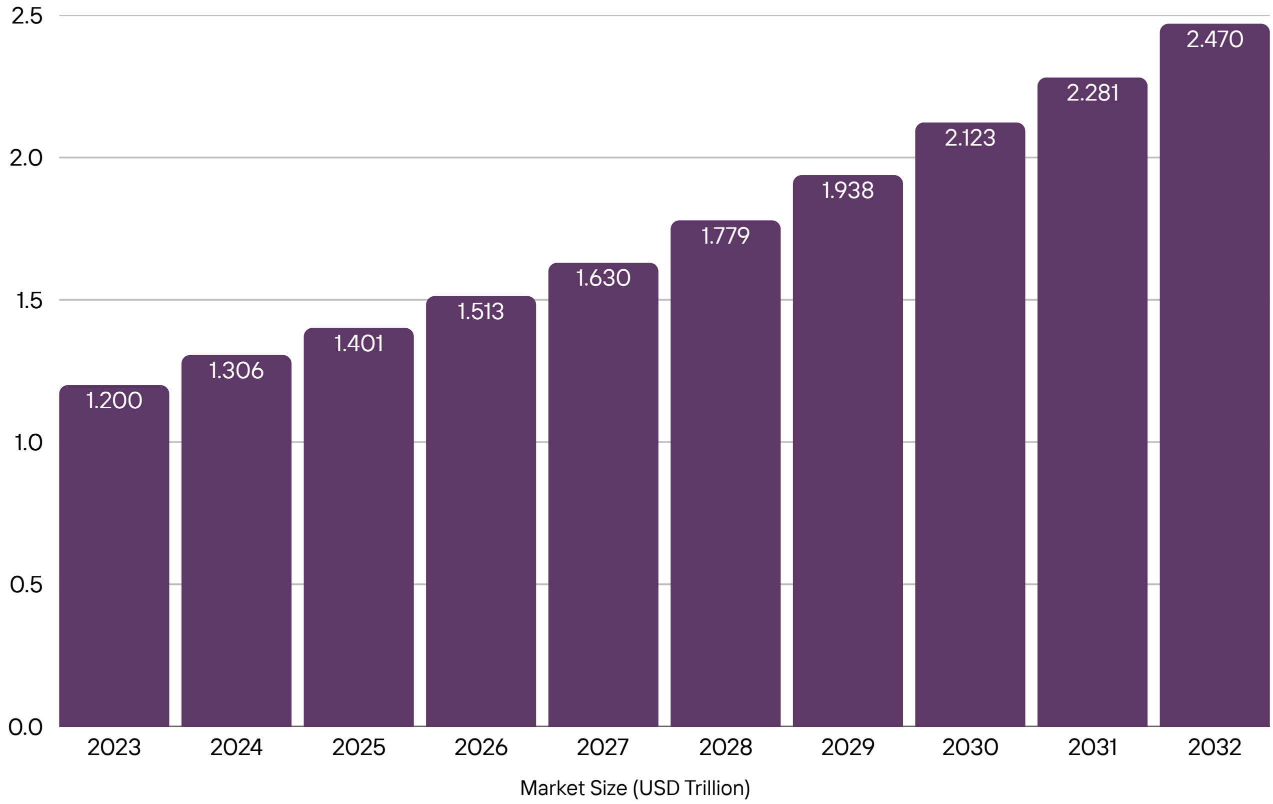
- **Risk Management:** Implementing models like Value-at-Risk (VaR) and stress testing to assess and mitigate financial risks.
- **Derivative Pricing:** Calculating the fair value of complex financial derivatives using models like Black-Scholes and Monte Carlo simulations.
- **Hedge fund management**, where quant funds drove **~22%** of global hedge assets and achieved **~8–11%** annual returns
- **Algorithmic Trading:** Designing automated trading strategies that execute orders based on predefined criteria and market conditions, with algo-based strategies accounting for **~85%** of US equity trading.
- **Portfolio Optimisation:** Allocating assets to maximise returns for a given level of risk, often using techniques like mean-variance optimisation.

Sources: https://gitnux.org/

Global Market Outlook

Market Size and Growth

As of 2023, the Global Quant Fund market is estimated to be valued at approximately USD 1.2 trillion. Projections indicate a Compound Annual Growth Rate (**CAGR**) of **8.5%**, anticipating the market to reach around USD 2.47 trillion by 2032.



Key Drivers

- **Technological Advancements:** The integration of Artificial Intelligence (AI) and Machine Learning (ML) has enhanced the capabilities of quantitative models, allowing for more sophisticated analysis and decision-making.
- **Data Availability:** The proliferation of big data and alternative data sources has provided quantitative analysts with richer datasets for model development.
- **Institutional Adoption:** Institutional investors are increasingly leveraging quantitative strategies to optimise portfolios and manage risks effectively.

Sources: Bar Graph

Global Market Outlook

Global Trends

- **Machine learning & AI adoption:** Usage rose **~58%** (2019–22); with ML-driven strategies outperforming traditional models by ~20% in 2023.
- **Alternative & ESG data integration:** Used by ~80% of quant firms; ESG-focused strategies saw 25% inflows in 2022.
- **Quantum and blockchain experiments:** Over 60% of institutions exploring quantum computing; blockchain data usage grew 67% (2019–23).
- **DeFi/crypto & digital asset models:** crypto quant strategies and smart-contract-based algos are expanding.
- **Cloud infrastructure & XAI:** heavy cloud use and demand for explainable AI amid tightening regulation.

Sources: https://gitnux.org/

Quant Finance In India

Market Overview

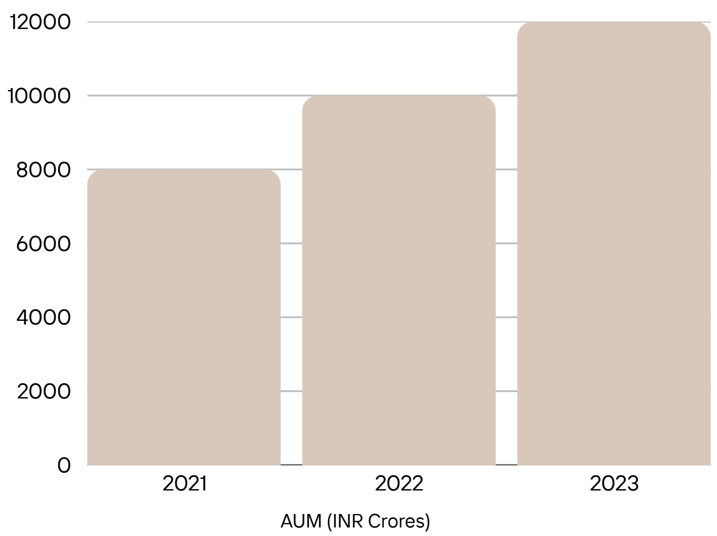
India's financial landscape is witnessing a gradual yet notable shift towards quantitative methodologies. While traditional investment approaches remain prevalent, the adoption of quantitative strategies is gaining momentum, particularly among institutional investors and fintech startups.

Current Penetration: Quantitative strategies currently account for approximately **1.5%** of the Indian mutual fund market. This is relatively modest compared to markets like the United States, where quant strategies represent over 35% of assets under management.

India's **algorithmic trading market** reached **$562 million in 2024**, with a projected CAGR of ~**9.5%** to reach **$1.27 billion by 2033**.
India's algorithmic trading is **~0.05% of global quant fund markets**, but its 9–10% CAGR matches global growth.

Growth Trends

- **Assets Under Management (AUM):** Quant-based mutual funds in India manage over **INR 12,000 crore**, reflecting a **50%** increase over the past three years
- **Trading Volume:** Quant strategies contribute to an estimated **25%** of trading volumes in India, up from 15% in 2019
- **Talent Pool:** India produces around 10,000 new professionals annually with expertise in data science and quantitative Finance, a number projected to grow by **20%** year-on-year



Sources: Bar Graph, https://www.imarcgroup.com/

Quant Finance In India

- **Increasing AI/ML platform integration**, infrastructure upgrades (co-location, cloud), and **the rise of HFT**.
- **More educational initiatives** (QuantInsti, IIQF) are driving talent inflow.

Government & Regulatory Stance

- **Algo trading was launched by SEBI in 2008**, enabling DMA, then refined with risk safeguards (kill-switches, audit trails).
- **SEBI sandbox** supports pilot fintech/quant innovations, processing within ~30 working days.
- **Feb 2025 rules:** Retail algo must be tagged, registered via exchange, and audited; algo trading accounted for 96–97% of prop/foreign profit on F&O in FY24.

Significance of Quant in India

Market Overview

- **Mutual funds:** Quant strategies have led to AUM growth (~₹12,000 crore).
- **Asset management & trading:** Automated execution, predictive risk controls, and retail participation have improved efficiency and reduced costs.
- **Fintech & robo-advisors:** Adoption of quant models in platforms like Zerodha, Upstox, and new retail offerings.
- **Proprietary trading firms:** Quant shops (e.g., iRage/QuantInsti tied firms) expanding globally; SEBI is auditing firms like Jane Street’s derivatives in India.

Sources: SEBI, reuters.com

Types of Quant Firms

Quantitative finance is implemented across a wide spectrum of financial institutions. These firms differ in their objectives, risk appetite, trading strategies, clients, and regulatory obligations. Below are the major categories:

Asset Management Companies

Asset managers invest on behalf of clients (institutional and retail) across asset classes, increasingly using quantitative models for portfolio optimisation, risk management, and passive strategies.

Global assets under management (AuM) are forecast to reach $145.4 trillion by 2025, up from $84.9 trillion in 2016 (6.2% CAGR). Active management is expected to account for $87.6 trillion (60%), passive management $36.6 trillion (25%), and alternatives $21.1 trillion (15%) by 2025.

Institutional money managers are responsible for investing funds on behalf of retail and institutional investors. These include mutual funds, pension funds, and ETFs.

Quant Integration:
- AMCs increasingly use quant models to construct portfolios, optimise risk/return tradeoffs, and automate rebalancing.
- Strategies include smart beta, factor investing, and tactical asset allocation.

Global Examples:
- BlackRock, Vanguard, State Street Global Advisors, Invesco QQQ
- These firms manage **trillions in AUM**, often with passive + smart-beta quant overlays.

Indian Examples:
- Nippon India Quant Fund, ICICI Prudential Quant Fund, Tata Quant Fund
- Total AUM in India’s quant-based mutual funds crossed **₹12,000 crore (~$1.45 billion)** by 2024, growing steadily.
- Quant is being used in passive funds, risk-based asset allocation models, and ESG score integration.

Proprietary Trading Firms (Prop Firms)

Prop trading firms use their own capital to trade financial instruments, employing quantitative models for market-making, statistical arbitrage, and high-frequency trading.

Key Features:

- No external clients or AUM obligations.
- Focus on absolute returns and fast execution.
- Heavy investment in low-latency infrastructure, co-location, and custom hardware.
- High secrecy and internal R&D teams to develop edge strategies.

Typical Strategies:

- Statistical arbitrage
- Market-making
- Momentum-based scalping
- Event-driven and latency arbitrage

Global Examples: Jane Street, Jump Trading, DRW, Citadel Securities, IMC, Hudson River Trading

Indian Examples:

- iRageCapital, Alphagrep, Dolat Capital, Samco, Five Star Trading
- India's proprietary trading is heavily involved in **index derivatives (Nifty, Bank Nifty)** and contributes up to **90% of volumes** in certain F&O segments.

Investment Banks

Large global investment banks maintain quant divisions for structuring products, executing trades, risk management, and pricing complex derivatives.

Functions:

- Quantitative research for options pricing, hedging, and fixed income analytics
- Trade execution algorithms (TWAP, VWAP, Implementation Shortfall)
- Risk systems design and stress-testing

Global Examples:

- Goldman Sachs, JP Morgan, Morgan Stanley, UBS, Barclays
- Many employ 1,000+ quants across trading, risk, and structuring teams.

In India:

- While full-scale quant desks are rare, quant roles exist in India-based GICs (Global In-house Centers) of JP Morgan, Morgan Stanley, and Goldman.
- Indian quants are increasingly hired into global roles from campuses like IIT Bombay, ISI, and IIMs.

Hedge Funds

Hedge funds pool capital from accredited investors to generate returns using a wide range of strategies, with quant hedge funds relying heavily on data-driven, systematic approaches.

The global hedge fund market is projected to grow from $4,971.75 billion in 2024 to $5,226.15 billion in 2025 (5.1% CAGR), and is expected to reach $6,019.79 billion by 2029.
Investment vehicles pooling capital from accredited investors to generate returns using various long/short, directional, or market-neutral strategies—often involving leverage and derivatives.

Key Features:
- May follow both **quantitative** and **discretionary** styles.
- Use of factor models, ML, and statistical systems.
- Performance-based fee structures (commonly 2% management + 20% performance fee).

Quant Hedge Fund Styles:
- Quant Equity (factor-based)
- Global Macro (ML-based forecasting)
- Volatility arbitrage
- Multi-strategy funds combining quant and discretionary layers

Global Examples:
- Renaissance Technologies (Medallion Fund)
- Two Sigma, DE Shaw, Man AHL, AQR Capital
- Renaissance's Medallion Fund has reportedly returned **over 66% annualised (net) between 1988–2018**.

Indian Examples:
- India has limited hedge funds due to regulatory barriers, but firms like Avendus Capital (Alternate Strategies) and Nine Rivers Capital run quant-heavy PMS products.
- Several offshore hedge funds (e.g., Millennium Management) now have India-based quant teams.

Quant Consulting and Data Analytics Firms

These firms provide specialised quant research, modelling, training, and data analytics services to hedge funds, banks, brokers, and fintech startups.

Offerings:
- Strategy research
- Portfolio risk diagnostics
- Data engineering for trading platforms
- Quant hiring and talent development
- Algorithm validation and compliance reviews

Global Examples:
- Numerai, WorldQuant, Coremont, Abel Noser
- Platforms like Quantopian (defunct) allowed open-access strategy building by retail quants.

Indian Examples:
- QuantInsti (via EPAT, iRage)
- International Institute of Quantitative Finance (IIQF)
- QuantSense, AlgoBulls provide analytics tools and custom strategies for brokers, HNIs, and traders.

Fintech Platforms and Robo-advisers

Startups and platforms use algorithmic engines to offer investment advice, trading signals, portfolio construction, and financial planning.
Core Characteristics:
- Scalable, automated, and data-driven
- Serve both retail and institutional clients
- Employ rule-based and ML-backed investment engines

Popular Models:
- SIP optimisation
- Risk scoring and asset recommendation
- Tax-loss harvesting
- Thematic and goal-based investing

Global Examples: Wealthfront, Betterment, SigFig, Acorns

Indian Examples:
- INDMoney, Groww, Zerodha's Smallcase, Kuvera, Scripbox
- Robo-advisors are expected to cross **₹2,500 crore AUM in India** by 2026, with growing retail trust in low-cost, quant-driven advice.

Trends and Innovations in Quantitative Finance

Quantitative Finance is experiencing a transformation driven by technological progress, data availability, and evolving market dynamics. Below are the major trends and innovations shaping the global and Indian quant ecosystem:

High-Frequency Trading (HFT)

HFT refers to ultra-fast, algorithm-driven trading strategies that execute thousands of trades per second, exploiting minute price inefficiencies.

Global Trends:
- Globally, HFT accounts for **50–60% of equity trading volume** in developed markets like the US and Europe.
- Major players include Citadel Securities, Jump Trading, Jane Street, and Virtu Financial.

In India:
- HFT comprises **40–45% of total trades** in Nifty and Bank Nifty futures on NSE.
- Enabled by Direct Market Access (DMA) and co-location services near exchanges, Indian HFT firms (like iRage, Alphagrep) are growing rapidly.
- SEBI has introduced latency audits and mandatory risk management tools (e.g., kill switches) to mitigate systemic risk.

AI/ML-Based Trading and Prediction Models

Overview: Machine Learning (ML) and Deep Learning (DL) models now enhance alpha generation, regime detection, volatility forecasting, and sentiment analysis.

Applications:
- **Supervised learning** (Random Forest, XGBoost) for price prediction.
- **Unsupervised learning** (clustering, PCA) for factor discovery.
- **Reinforcement Learning** (e.g., Deep Q-Learning) for trade execution and portfolio rebalancing.
- **Natural Language Processing (NLP)** is used in news analytics, social media signals (Twitter, Reddit), and earnings call analysis.

Impact:
- AI-driven funds like Two Sigma and Renaissance Technologies have outperformed traditional hedge funds.
- In India, quant teams at Zerodha, small fintech firms, and prop desks are adopting ML for signal generation and risk assessment.

Crypto Quant Strategies (including DeFi Subset)

Rise of Crypto Quants:

- Quant strategies now play a significant role in the **$1.8 trillion** global crypto market.
- Strategies include statistical arbitrage, mean reversion, liquidity mining, and basis trading.

DeFi Subset:

- Decentralised Finance (DeFi) allows algorithmic strategies across DEXs (e.g., Uniswap, Curve) without intermediaries.
- Strategies include flash loan arbitrage, impermanent loss hedging, and yield optimisation.
- Protocols like **dYdX** and **GMX** are attracting quant developers with on-chain APIs and low-latency execution.

India Scene:

- While crypto regulations are tight, Indian quants participate via offshore accounts or work with international firms.
- India's **WazirX** and **CoinDCX** are exploring ML-backed analytics tools, though still in infancy.

No-Code Quant & Strategy Builders

No-code platforms allow traders and researchers to build, backtest, and deploy strategies without writing code.

Popular Global Tools:

- **QuantConnect**, **Kavout**, **AlgoTrader**, and **Trality**.
- Drag-and-drop interfaces combined with broker integration (Interactive Brokers, Alpaca, Zerodha API).

India-Specific:

- **QuantPower**, **Streak (by Zerodha)**, and **Algobulls** are leading the no-code algo revolution.
- These platforms have enabled non-programmers (e.g., retail investors, finance students) to engage in live algorithmic trading with risk checks and SEBI compliance.

Automated Portfolio Management & Robo-Advisors

Robo-advisors use rule-based or ML-enhanced algorithms to provide financial planning, portfolio allocation, and rebalancing without human intervention.

Global Scene:
- Vanguard, Betterment, and Wealthfront collectively manage over **$500 billion** in AUM.
- Portfolios are constructed using Modern Portfolio Theory (MPT), Risk Parity, and Factor Investing.

In India:
- **INDMoney**, **Groww**, and **Smallcase** are leading robo-advisor and semi-automated portfolio platforms.
- They combine traditional advice with quant models for SIPs, ETF allocations, and tax harvesting.
- A 2024 IAMAI report estimated **robo-managed AUM in India to surpass ₹1,200 crore**, growing at over **30% CAGR**.

Quantitative Research & Tools

Quantitative Finance is experiencing a transformation driven by technological progress, data availability, and evolving market dynamics. Below are the major trends and innovations shaping the global and Indian quant ecosystem:

Open-Source Tools & Platforms

- **QuantLib**, **Backtrader**, **Zipline**, **PyAlgoTrade**, **TensorTrade**, etc.
- Platforms like **QuantInsti's EPAT** provide structured development.

Datasets & APIs

- **Quandl**, **Bloomberg**, **Reuters**, and exchanges' data feeds.
- Indian brokers/fintech firms now offer **data & model APIs**, with SEBI-approved sandboxes.

Model Validation & Risk Management

- Emphasis on backtesting, stress-testing, live vs simulated performance.
- SEBI mandates risk controls: kill-switches, pre-trade risk checks, algo tagging, and audit trails.

Our Competitors

- BacktestingMax
- QuantConnect
- QuantInsti
- TradingView
- TrendSpider
- Amibroker
- QuantPedia
- WorldQuant BRAIN
- MetaTrader 4
- Backtrader
- Zipline

BacktestingMax

Vision:
To give traders the ability to test trade ideas and refine their skills cost-effectively.

Features:
Unlimited trades, TradingView integration, multiple markets like forex pairs, indexes, and crypto pairs, all timeframes, detailed overviews reflecting Sharpe ratio, profit, equity curve, maximum drawdown, bar replay to jump through the chart, easy lot size calculation (you just need to select your preferred risk).

Pricing:
Free version: Forex and index trades, basic drawing tools, unlimited backtesting sessions
Premium at 10$/month: Over 500 crypto pairs, all drawing tools, indicators, economic calendar, unlimited data range, unlimited AI assistant access, AI chart analysis, and quick support.
Lifetime premium at $150.

Partnership:
They provide special discounts and allot referral codes to subscribers. Subscribers receive commission if someone uses their code to sign up.

Reviews and stats:
Rated 4.3 on Trustpilot. Has 38000+ users and 700+ subscribers. It has been running for 1.5 years now.

Sources: https://backtestingmax.com/

QUANTCONNECT

Vision:

Target customers:

They are mainly targeting researchers, teams, trading firms and institutions. They offer separate plans for each type of user.

Pricing:

Free plan includes: Equity, forex, crypto, futures, Options data, community support and unlimited backtesting.
Researcher plan at $60/month: Local coding, expanded datasets access, unlimited projects, large files, add up to 2 compute nodes.
Team plan at $120 per user/month - Ideal for start ups. It offers project collaboration, increased log, File limits, users can add up to 10 compute nodes.
Trading firm plan at $336 per user/month: It offers team project ownership, permissions management, it allows its users to add unlimited compute nodes.
Institution plan at $1080 per user/month: On top of everything provided by the firm plan, it include s an on-premise platform, AES-256 Code encryption, FIX/professional brokerages.

Supported brokerages:

InteractiveBrokers, coinbase, BITFINEX, tradier brokerage, TRADING TECHNOLOGIES, Bloomberg, BINANCE, Kraken, Terminal Link, WEX, charles SCHWAB, SS&C, SAMCO, Zerodha.

Reviews and stats:

They have formed a community of 392K users, with 500K+ backtests per month. Their trading volume is $45 B per month. The returns over market have been above 7%
Total equity fucding - $2.46M.
Total funding - $84.2 M
It is ranked 4th among its 180 competitors..
It holds 13% of the market share.

Sources: https://www.quantconnect.com/

QuantInsti

Founded in 2010 as the educational arm of iRage, one of Asia's leading high-frequency trading firms, QuantInsti has emerged as a global leader in algorithmic and quantitative trading education and technology. Headquartered in Mumbai, India, the company has built a strong international presence, engaging over 1 million users annually across 190+ countries. Its mission is to democratize access to knowledge and tools in algorithmic trading, serving both individual traders and institutional clients.

Business Model & Key Offerings

Flagship Program: EPAT

- Executive Programme in Algorithmic Trading (EPAT) is a 6-month, fully online, instructor-led certification program.
- Recognized globally as the first proctored certification in the domain.
- Developed by 20+ industry veterans, the curriculum spans strategy design, execution, risk management, and business setup.
- Accredited by the Institute of Banking & Finance (IBF) Singapore.

Quantra

- An interactive, self-paced learning platform focused on Python and algorithmic trading.
- Offers 60+ hands-on courses across asset classes and trading strategies.
- Seamless integration with Blueshift for practical implementation and strategy testing

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Blueshift

- An institutional-grade, cloud-based platform for research, backtesting, and live trading.
- Among the few public-facing Complex Event Processing (CEP) engines available in India.
- Supports multi-asset, multi-market strategies.

Corporate Solutions

- Trusted by 300+ corporates in over 20 countries.
- Offers customized edtech and fintech solutions to brokerage firms, hedge funds, and other financial institutions.

Sources: https://www.quantinsti.com/

Free Learning Resources

- Includes blogs, eBooks, webinars, and courses.
- Designed to foster community engagement and ongoing professional development.

Market Position & Global Reach

- User Base: 1M+ active users per year, with over 60% opting for multiple courses.
- Customer Satisfaction: More than 13,000 five-star reviews across platforms.
- Geographic Penetration: Users in 190+ countries, with key markets in Asia, North America, the UK, and MENA.
- Corporate Partnerships: 300+ enterprise clients, including top-tier financial institutions and brokers.
- Career Services: Lifetime placement support for EPAT alumni, with notable placements offering salaries upwards of ₹50 lakhs in India.

Financial Highlights (2025 Estimates)

- Annual Revenue: $17.9 million
- Team Size: 128 employees
- Revenue Per Employee: Approx. $140,000

Sources: https://www.quantinsti.com/

TradingView

TradingView's vision is to democratize access to financial information and empower individuals globally to take control of their financial future. TradingView stands out for its user-friendly interface, community-driven content, and extensive charting capabilities, making it a preferred choice for both novice and experienced traders.

- **Global Reach:** TradingView ranks among the top 130 websites globally, reflecting its widespread adoption and user base.
- **User Engagement:** The platform is widely used by retail traders, professionals, and institutions for market analysis, backtesting, and collaborative research.
- **Financial Performance:** While specific revenue figures are not publicly disclosed, TradingView has demonstrated strong growth through multiple funding rounds.
- **Funding and Valuation:** TradingView has raised a total of $339.37 million across six funding rounds. Its latest Series C round in October 2021 raised $298 million, valuing the company at $3 billion.
- **Investors:** Notable investors include Insight Venture Partners, DRW Venture Capital, Jump Capital, and Tiger Global Management.

The platform is used in over 190 countries, with major user bases in India, the United States, Russia, Turkey, and Japan. Coverage includes stocks, indices, forex, cryptocurrencies, commodities, and futures from over 100 global exchanges.

Features:

- **Advanced Charting:** Offers real-time and historical price charts with a wide array of technical analysis tools.
- **Social Trading:** Enables users to publish trading ideas, follow top analysts, and engage in community discussions.
- **Multi-Asset Coverage:** Supports analysis of stocks, forex, cryptocurrencies, indices, and commodities.
- **Customizable Alerts:** Users can set alerts for price movements, technical patterns, and news events.
- **Broker Integration:** TradingView integrates with numerous brokers, allowing users to execute trades directly from the platform.

Sources: https://www.tradingview.com/

Target Customers

TradingView primarily targets:

- **Individual and professional traders:** Retail investors, day traders, swing traders, and technical analysts who require advanced charting and social features.
- **Institutional investors:** Hedge funds, asset managers, and professional trading desks seeking robust analytics and data visualization.
- **Financial media and websites:** Platforms that embed TradingView's charts and widgets for their own audiences.
- **Developers and third-party vendors:** Those creating custom indicators, trading tools, or integrating TradingView's technology into other platforms.

Sources: https://www.tradingview.com/

Pricing:
TradingView offers a range of subscription plans designed to cater to different user needs, from casual traders to professional analysts.

- The platform provides a free tier with basic features, including one chart per tab, two indicators per chart, and five price alerts.
- Essential plan at $16.95 per month, which removes advertisements and increases limits to two charts per tab, five indicators per chart, and 20 price alerts.
- Plus plan, priced at $33.95 per month, further expands these limits to four charts per tab, ten indicators per chart, and 100 price alerts.
- Premium plan, at $67.95 per month, offers eight charts per tab, 25 indicators per chart, and 400 price alerts.
- Expert plan is available at $199.95 per month, with ten charts per tab, 30 indicators per chart, and 600 price alerts.
- Ultimate plan costs $239.95 per month and provides 16 charts per tab, 50 indicators per chart, and 1,000 price alerts.

Annual subscriptions come with a discount equivalent to two free months. Additional fees may apply for real-time data from specific exchanges and for API access.

Sources: https://www.tradingview.com/

TrendSpider

TrendSpider is designed for both beginner and advanced traders, automating routine analysis and enabling users to focus on strategy and execution. The platform is recognized for its robust idea generation, market scanning, and powerful alerting system, helping traders uncover and act on opportunities efficiently.

Global Reach & User Base:
TrendSpider serves tens of thousands of active traders globally, with a distributed team across North America, Europe, and Asia. The platform is used by a wide spectrum of market participants, including retail traders, professional analysts, and institutional investors. Its user base spans individual traders, ETF operators, family offices, financial advisors, wealth managers, hedge funds, and portfolio managers.

Market Coverage:

- **Asset Coverage:** Over 118,000 US assets, 43,000 global assets, and 73,000 alternative data feeds.
- **Data Feeds:** 94,000+ real-time, 84,000+ delayed, and 11,000+ periodical feeds.
- **Instruments:** Stocks, ETFs, indices, forex, cryptocurrencies, futures, and commodities.
- **Historical Data:** Up to 50 years of historical data for backtesting and research.

Pricings:
TrendSpider typically offers a 7-day free trial for new users.

- **Essential Plan:** $39/month (billed annually) or $49/month (billed monthly)
- **Elite Plan:** $79/month (billed annually) or $99/month (billed monthly)
- **Advanced/Professional Plan:** $129/month(annually) or $149/month(monthly)
- **Professional Market Data:** For professional users, market data access is an additional $20/month.
- **Discounts:** Lightspeed Connect API users can access the Enhanced plan at a 69.5% discount.

Sources: https://trendspider.com/

Target Customers:
TradingView primarily targets:

- **Active individual traders:** Day traders, swing traders, and technical analysts seeking automation and advanced charting.
- **Quantitative and algorithmic traders:** Users who need robust backtesting, custom scripting, and AI-driven strategy development.
- **Long-term and fundamental investors:** Those who require deep data, alternative data feeds, and fundamental overlays.
- **Institutions and professionals:** Hedge funds, RIAs, portfolio managers, and research desks.
- **Developers:** Users building custom indicators, bots, or integrating the platform with other tools.

Sources: https://trendspider.com/

Features:
TrendSpider distinguishes itself through a suite of advanced, automation-focused tools:
Automated Technical Analysis: AI-powered detection of trendlines, patterns, Fibonacci levels, and support/resistance zones.
Smart Dynamic Alerts: Alerts that adapt dynamically to price and indicator changes, not just static price levels.
Multi-Timeframe Analysis: Analyze and overlay multiple timeframes simultaneously for deeper trend confirmation.
Automated Pattern Recognition: AI identifies chart patterns such as triangles, head & shoulders, and double tops in real time.
Backtesting & Strategy Development: Visual, no-code, and JavaScript-based strategy creation and backtesting with up to 50 years of data.
Raindrop Charts: Unique volume-based price visualization tool for enhanced market sentiment analysis.
AI Strategy Lab: Train custom predictive models and machine learning strategies directly on the platform.
Custom Indicators & Studies: Build and deploy custom indicators using JavaScript.
Broker Integration: Direct trading via select brokers and automated execution through SignalStack.

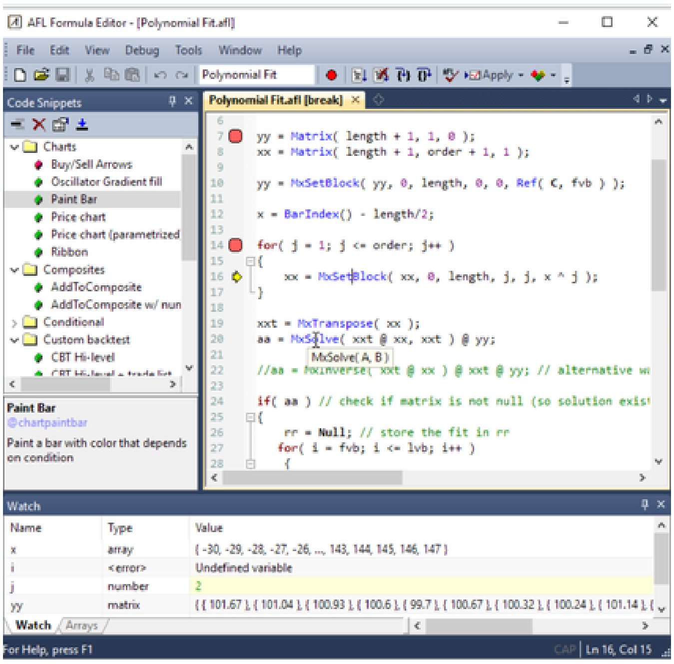
Sources: https://trendspider.com/

AmiBroker

It is a trading company based in Poland, founded in 1995. It has not raised any funds yet. It provides technical analysis solutions used by traders to help with their trading. The code needs to be written in AFL (AmiBroker Formula Language),
Competitors: 37 active, out of which 1 is funded and 1 has quit.

Products: AmiBroker, AmiQuote, AFL Code Wizard-

- AmiBroker is a backtesting platform.
- Amiquote automates the process of downloading data from various online sources and imports it into AmiBroker.
- AFL Code Wizard is an AI product where we can type in strategies in English to receive code in AFL.

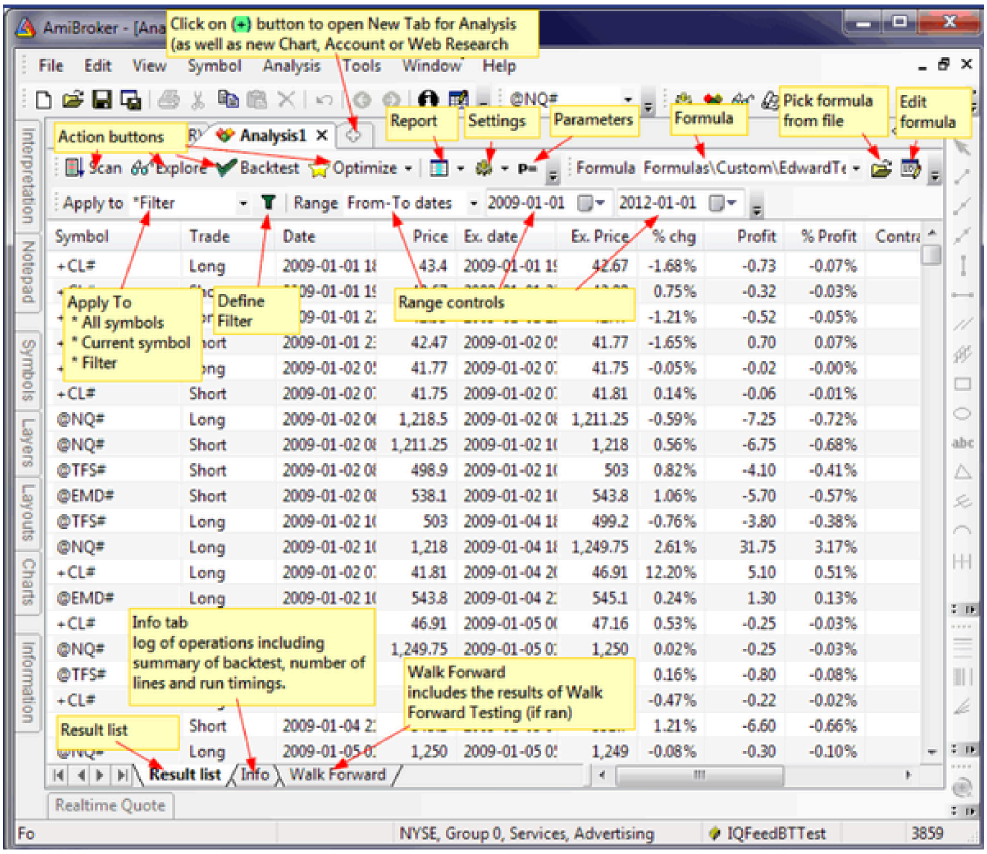


The AFL formula editor window

Services: Charts - Drag-and-drop averages, bands and indicators on other indicators, modify parameters in real-time using sliders and customize using many different styles & gradients to make them beautiful.
Portfolio backtesting and optimization - advanced position sizing, scoring and ranking, rotational trading, custom metrics, custom backtesters, multiple-currency support
Automation and batch processing: We can run it from Windows scheduler. This is a new feature.

Sources: https://www.amibroker.com/

Their website shows all the features they have included.



This is the Analytics window, where all the scans, explorations, portfolio backtests, optimizations, walk forward tests along with Monte Carlo simulation are visible.

Additional features:
Uses AI optimization (Particle Swarm and CMA-ES) to search huge spaces in limited time, allowing us to try thousands of different parameter combinations to find best performing ones.
Built in debugger, the code editor has syntax highlighting, auto-complete, parameter call tips, code folding, auto-indenting and in-line error reporting. When you encounter an error, meaningful message is displayed right in-line so you don't strain your eyes. Ready to use code snippets are available.
Multi-threading: All your formulas automatically benefit from multiple processors/cores. Each chart formula, graphic renderer and every analysis window runs in separate threads.

Sources: https://www.amibroker.com/

Pricing:
Offers a 30 day free trial, but does not have a free version.

Standard Edition($299)
; Entry-level version for End-of-day and swing traders. End-of-day and Real time. Intraday starting from 1-minute interval. 10 symbols limit in Real time Quote window. 2 simultaneous threads per Analysis window. 32-bit only.

Professional Edition($ 369):
Professional Real-Time and Analytical platform with advanced backtesting and optimization. End-of-day and Real time. All Intraday Tick/Second/Minute intervals, Unlimited symbols in Real time Quote window. Unlimited symbols in Time&Sales. MAE/MFE stats included. Up to 32 simultaneous threads per Analysis window. Includes both 64-bit and 32-bit versions.

Ultimate Pack Pro ($499):
Everything that AmiBroker Professional Edition has plus two very useful programs:
AmiQuote - quote downloader from multiple on-lines sources featuring free EOD and intraday data and free fundamental data.
AFL Code Wizard - creates AFL formulas out of plain English sentences. Invaluable learning tool for novices.

AmiQuote and AFL Code Wizard licenses are worth $198 when purchased separately.

All these packages come with 24 months of free ungrades and support.

Valuation:
Seed - $14.97M
Series A - $42.8M
Series B - $180M

Investors:
4 Venture Capitalists: 4growth VC, Digital Ocean Ventures, FundingBox Deep Tech fund, Arkley
2 incubators: FundingBox, Arkley
Other: nChain

Sources: https://www.amibroker.com/

QuantPedia

Quantpedia provides financial academic research in a user-friendly form to help anyone who seeks new quantitative trading strategy ideas. The company evaluated quantitative research papers based on the strategy's implementability, backtest length period, and overall soundness. Selected strategies are then added to the existing Quantpedia structure. The company extracts trading rules in plain language, performance and risk characteristics, and various descriptive characteristics. Each strategy is then categorized by several keywords, and database connections to similar strategies are created.

So, mainly they provide strategies.
Process: First, they take data from research portals, finance journals, universities, and conferences. Then they investigate the data and keep it if it is relevant.

Pricing:
Prime ($299 for 3 months): Focuses on tactical asset allocation and market timing strategies, suitable for individual investors and beginners.
Premium ($449 for 3 months): Unlocks the full strategy database and advanced screener, ideal for quants and active traders.
Pro ($599 for 3 months): Adds sophisticated portfolio construction, custom reporting, and deep analytics for professionals and institutions

Strengths:

- Bridges theory and practice effectively.
- Constantly updated with new research.
- Clean UI with detailed categorization.
- Great for idea generation and portfolio strategy mix.

Downsides:

- Not all strategies are easily tradable in retail environments.
- Full access requires a paid plan.
- Some backtests are based on academic assumptions (e.g., no slippage or transaction costs).

Market performance:
It holds 9% of the total market among its competitors.
Its competitors are: Composer, Quantconnect, AlgoBulls, and Streak.
It has received funding of $84.2M

Sources: https://quantpedia.com/

WorldQuant BRAIN

The platform aims to harness the collective intelligence of its users to generate predictive algorithms (“alphas”) and advance the application of AI and machine learning in finance. WorldQuant seeks to scale the number of algorithms deployed across its business by engaging global talent and leveraging innovative technology, with the ultimate goal of delivering superior results in quantitative investment.

Global Reach & User Base:
Over 100,000 users participate on the BRAIN platform. The BRAIN platform is accessible globally, with active users and consultants from over 100 countries.

Market Coverage:

- Asset Classes: BRAIN focuses on equities, but also provides data and modeling capabilities across a broad range of asset classes and global markets.
- Data Fields: The platform offers access to over 120,000 data fields for research and model development.
- Simulation Environment: Users can leverage WorldQuant’s proprietary simulation tools to backtest and refine their algorithms using deep historical market data.
- Alpha Factory: WorldQuant’s broader infrastructure includes a repository of over 4 million alphas developed from thousands of data sets, which are used to construct investment strategies.

Pricing:

- Participation: Access to the BRAIN platform and its competitions is free for users.
- Compensation: Top-performing participants may be invited to the BRAIN Research Consultant Program, where they are paid for their research contributions.
- No Subscription Fees: There are no subscription or licensing fees for individual users; the model is based on crowdsourcing and performance-based compensation.

Features:

- Interactive Simulation Platform: Users build and test alphas in a real-world-like environment.
- Crowdsourcing Model: Anyone with quantitative skills can contribute, compete, and potentially become a paid consultant.
- Professional Development: The platform offers learning resources and opportunities to apply AI and machine learning concepts in finance.
- Global Collaboration: Consultants work remotely and collaborate with WorldQuant's internal teams on research projects.
- Performance Tracking: Dashboards and analytics help users monitor and improve their contributions.

Target Customers:

- Quantitative Researchers & Data Scientists: Individuals with strong analytical, programming, and mathematical skills, including students and professionals seeking to break into quant finance.
- Aspiring and Experienced Quants: Users interested in developing and testing financial algorithms and gaining professional experience.
- Global Talent: The platform is open to users from around the world, with specific consultant opportunities in select countries.
- Academic Community: Students and researchers seeking hands-on experience in quantitative finance and AI applications.

MetaTrader4

Global Reach & User Base:
MT4 is one of the most widely used trading platforms in the world, with millions of traders across more than 100 countries. Its user base includes retail traders, professional forex traders, and institutions. The platform's popularity is driven by its reliability, extensive broker support, and a large community of developers and users.

Market Coverage:

- Asset Classes: MT4 is primarily focused on Forex trading but also supports CFDs, indices, commodities, and other derivatives, depending on broker integration.
- Broker Integration: The platform is offered through hundreds of brokers worldwide, each providing access to various markets and instruments.
- Mobile and Desktop Access: Available on Windows, iOS, and Android, allowing for seamless trading and analysis across devices.

Features:

- Advanced Technical Analysis: Dozens of built-in indicators, charting tools, and timeframes for comprehensive market analysis.
- Flexible Trading System: Supports multiple order types, instant and pending orders, and one-click trading.
- Algorithmic Trading: Users can develop, test, and deploy automated trading strategies (Expert Advisors) using the built-in MQL4 programming language.
- Trading Signals: Allows users to subscribe to and automatically copy the trades of successful traders.
- Marketplace: Access to a vast library of paid and free Expert Advisors, indicators, and scripts.
- Mobile Trading: Full-featured trading apps for smartphones and tablets.

Sources: https://www.metatrader4.com/en

Pricing:

- Platform Access: MT4 is free for end users; costs are typically borne by brokers who license the platform from MetaQuotes.
- Expert Advisors and Indicators: Users can purchase or download free Expert Advisors and technical indicators from the MT4 Market. Prices for these add-ons vary, with some being free and others ranging from a few dollars to several hundred dollars.
- Trading Signals: Subscription fees for copying trading signals are set by signal providers and can vary widely.
- Broker Fees: Users may incur spreads, commissions, and other trading fees depending on their chosen broker.

Target Customers:

- **Retail Forex Traders:** Individuals seeking a reliable, feature-rich platform for manual and automated trading.
- **Professional Traders:** Those requiring advanced charting, algorithmic trading, and custom strategy development.
- **Brokers and Financial Institutions:** Companies offering trading services to clients.
- **Developers:** Programmers creating custom indicators, scripts, and automated trading systems for the MT4 ecosystem.

Sources: https://www.metatrader4.com/en

Backtrader

Global Reach & User Base:

- Individual traders and quant enthusiasts seeking to test and automate their strategies.
- Quantitative developers and data scientists in need of a flexible, programmable backtesting engine.
- Small to medium-sized trading firms and research teams looking for a customisable, cost-effective solution for strategy research and live deployment.

Market Coverage:

- **Asset Classes:** Originally designed for equities and forex, Backtrader also supports derivatives, commodities, and cryptocurrencies.
- **Data Feeds:** Compatible with a wide range of data sources, including CSV files, databases, and real-time broker feeds (e.g., Interactive Brokers, OANDA, Alpaca).
- **Broker Integration:** Enables live trading through integration with several popular brokers, allowing seamless transition from backtesting to live execution.
- **Multi-Asset & Multi-Timeframe:** Users can test strategies across multiple instruments and timeframes simultaneously.

User Engagement:

Backtrader's open-source model and Python foundation encourage high user engagement through:

- Community-driven development and support.
- Extensive sharing of strategies, indicators, and analyzers.
- Active forums, blogs, and third-party tutorials.

Pricing:

- Open Source & Free: Backtrader is free to use for all users.
- Potential Costs: Users may incur expenses when integrating with live brokers (such as broker commissions or API fees) or when purchasing premium historical data feeds.

Sources: https://www.backtrader.com/

Features:

- **Backtesting Engine:** Simulate trading strategies on historical data to evaluate performance, risk, and robustness.
- **Custom Indicators & Strategies:** Easily create and test proprietary indicators and strategies using Python.
- **Parameter Optimisation:** Built-in support for optimising strategy parameters to maximise performance.
- **Automated Trading:** Deploy strategies for live trading with broker integration, supporting automated execution and risk management.
- **Risk Management Tools:** Includes stop-loss, take-profit, and position sizing features.
- **Extensive Documentation:** Comprehensive guides, examples, and API references for all levels of users.

Target Customers:

- Individual traders and hobbyists seeking to experiment with and refine trading strategies.
- Quantitative analysts and developers who require a programmable, flexible backtesting and trading engine.
- Small and medium-sized trading firms looking for a customizable, budget-friendly research and deployment platform.
- Academic researchers and students in quantitative finance and data science.

Sources: https://www.backtrader.com/

Zipline

Zipline is a widely used, open-source Python library and platform for backtesting and executing algorithmic trading strategies. Originally developed by Quantopian in 2012, Zipline became the backbone of Quantopian's $100 million crowd-sourced hedge fund and remains a core tool in the quantitative trading community, even after Quantopian's closure.

Global Reach & User Base

- **Open-source adoption:** Zipline is widely used by individual quants, academic researchers, and small trading firms globally.
- **Community-driven:** Its user base spans hobbyists, students, professional developers, and quant researchers, supported by an active online community and public forums35.
- **Integration:** Zipline was the core backtesting engine behind Quantopian, a platform that enabled thousands of users to develop and test trading strategies in the cloud26.

Key Features

- **Event-Driven Backtesting:** Simulates realistic trading environments, including slippage, commissions, and order delays, to avoid look-ahead bias and ensure robust results29.
- **Pythonic API:** Designed for ease of use, with strategy logic written in Python and seamless integration with popular data science libraries (Pandas, NumPy, matplotlib)2.
- **Custom Indicators & Strategies:** Users can create, test, and optimise custom indicators and strategies using Python.
- **Parameter Optimisation:** Built-in support for optimising strategy parameters to improve performance.
- **Risk Management Tools:** Includes support for stop-loss, take-profit, position sizing, and risk metrics like Sharpe ratio.
- **Extensive Documentation:** Offers comprehensive guides, tutorials, and API references for users at all levels3.
- **Community Extensions:** Projects like Zipline-Live and Zipline-Trader extend functionality for live trading and support for additional brokers and data sources8.

Sources: https://github.com/quantopian/zipline

Market Coverage

- **Asset Classes:** Primarily designed for equities and ETFs, but also supports futures, forex, and cryptocurrencies through extensions and third-party integrations2.
- **Data Sources:** Compatible with a wide range of data feeds, including Quandl, Yahoo Finance, CSV files, and live brokerage APIs (e.g., Interactive Brokers, Alpaca)2.
- **Broker Integration:** Supports live trading through integration with brokers like Interactive Brokers and Alpaca, especially via projects such as Zipline-Live and QuantRocket8.
- **Multi-Asset & Multi-Timeframe:** Capable of handling multiple assets and timeframes within a single strategy.

Pricing

Zipline is completely free to use for all users under an open-source license3.

Target Customers

Individual quants and hobbyists: People seeking a flexible, programmable backtesting engine for strategy development.
Academic researchers and students: Those studying quantitative finance, data science, or algorithmic trading.
Small trading firms and research teams: Organizations looking for a customisable, cost-effective research and deployment platform.
Developers: Programmers building custom tools, indicators, or integrating Zipline into larger trading systems.

Sources: https://github.com/quantopian/zipline